

Math Camp Exam

Columbia Economics PhD Math Camp

August 29th, 2025

This is a timed remote exam. You have time until noon (12pm) to submit your answers via email to the TA (hj2688@columbia.edu) and cc the instructor (be2329@columbia.edu). There are 6 questions for a total of 100 points, plus 15 bonus points.¹ No questions by email are allowed. Please justify your answers, partial credit will be awarded in case of incomplete answers, and no credit is awarded for a correct answer with no procedure shown. The exam is closed book and notes, and you are not allowed to consult peers or use the web to aid in solving the problems. A calculator is allowed. Columbia's academic honor code applies.

1. (15 points) Let $\mathcal{C}_{[0,1]} := \{f : [0, 1] \rightarrow \mathbb{R} : f \text{ is continuous}\}$ and $d_\infty : \mathcal{C}_{[0,1]} \times \mathcal{C}_{[0,1]} \rightarrow \mathbb{R}_+$ be defined as

$$d_\infty(f, g) := \max_{t \in [0,1]} |f(t) - g(t)|.$$

- (a) (2 pts) For $f, g, h : [0, 1] \rightarrow \mathbb{R}$ with $f(x) = x$, $g(x) = x^2$ and $h(x) = 0.5x^2$, find $d_\infty(f, g)$ and $d_\infty(f, h)$.
- (b) (3 pts) Argue that d_∞ is a well-defined function. (Hint: You can use as fact that the absolute value is a continuous transformation.)
- (c) (10 pts) Show that $(\mathcal{C}_{[0,1]}, d_\infty)$ is a metric space.
2. (25 points) Let $\rho : \mathbb{R} \times \mathbb{R} \rightarrow \{0, 1\}$ be the discrete metric, i.e.

$$\rho(x, y) := \begin{cases} 1, & \text{if } x \neq y \\ 0, & \text{if } x = y \end{cases}$$

- (a) (5 pts) Show that any singleton in $\mathbb{R}$ (i.e. any set made up of a single point $\{x\} \subset \mathbb{R}$) is open in $(\mathbb{R}, \rho)$.
- (b) (5 pts) Show that any subset of $\mathbb{R}$ is open in $(\mathbb{R}, \rho)$.
- (c) (5 pts) Show that any subset of $\mathbb{R}$ is closed in $(\mathbb{R}, \rho)$.
- (d) (5 pts) Show that any subset of $\mathbb{R}$ is bounded in $(\mathbb{R}, \rho)$.
- (e) (5 pts) Show that $[0, 1]$ is not compact in $(\mathbb{R}, \rho)$.
- (f) (**Bonus:** 5 pts) Show that the sequence $(1/n)$ does not converge to 0 in $(\mathbb{R}, \rho)$.

¹Bonus points are added to your exam's total score as long as it does not go over 100. For example, if you score 96 in the main questions and 8 in the bonus questions, your total score will be capped at 100. You should only attempt bonus points if you have extra time or want attempt the bonus questions instead of some of the main questions.

3. (15 points) Define $\mathbf{1}_n \in \mathbb{R}^n$ as a column vector of size n full of ones, i.e. $\mathbf{1}_n := (1, 1, \dots, 1)$. Now define $\mathbf{S}_n := \mathbf{1}_n \mathbf{1}_n^T$, for any $n \in \mathbb{N}$.

- (a) (3 pts) Find the eigenvalues of $\mathbf{S}_2$ and $\mathbf{S}_3$.
- (b) (7 pts) Find $\sigma(\mathbf{S}_n)$ for $n > 3$. (Hint: Find a pattern from your answer to part (a), and use row operations that preserve determinant.)
- (c) (5 pts) In general, is $\mathbf{S}_n$ diagonalizable in $\mathbb{R}$? If so, propose a spectral decomposition; that is, find an invertible matrix P and diagonal matrix D such that $A = PDP^{-1}$. (Note: You do *not* have to compute P^{-1} .)

Next, a bonus question about vectors unrelated to (a)-(c):

- (d) (**Bonus:** 5 pts) If $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are L.I. non-zero vectors in some vector space $(V, +, \cdot)$, is $\{\mathbf{v}_1, 2\mathbf{v}_1 + \mathbf{v}_3, 4\mathbf{v}_2 + 2\mathbf{v}_3\}$ a basis of $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$?
4. (10 points) Define $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ with

$$f(u_1, u_2, u_3) = \begin{bmatrix} \ln(u_1 + u_2^2 + u_3^3) \\ u_1^2 \end{bmatrix}, \quad g(v_1, v_2) = \begin{bmatrix} e^{v_1} \\ \sqrt{v_2} \end{bmatrix}, \quad h(w_1, w_2) = w_1^2 - w_2^2.$$

- (a) (2 pts) Compute the Jacobian matrices of f , g and h .
- (b) (3 pts) Compute the partial derivatives:

$$\frac{\partial h \circ g \circ f}{\partial u_i}$$

for $i = 1, 2, 3$.

- (c) (5 pts) Approximate the value of $h \circ g \circ f(1.1, 1.9, -0.8)$ using a first order Taylor approximation. (Hint: Use (1,2,1) as reference point).
 - (d) (**Bonus:** 5 pts) Approximate the value of $h \circ g \circ f(1.1, 1.9, -0.8)$ using a second order Taylor approximation.
5. (15 points) Find the maximizers of the following problem

$$\max_{(x,y,z) \in \mathbb{R}^3} f(x, y, z) = -x^2 + 2x - y^2 + 4y - z^2 + 6z \quad \text{subject to} \quad x \leq 2, y + 2z \geq 12$$

6. (20 points) Let $(X, d_X), (Y, d_Y)$ be metric spaces. We say $F : \mathbb{R} \rightrightarrows Y$ has a compact graph iff the graph of F is a compact set in metric space $(X \times Y, d_{X \times Y})$ where

$$d_{X \times Y}((x, y), (x', y')) := \sqrt{d_X(x, x')^2 + d_Y(y, y')^2}$$

Use as given the fact that x_n converges to x in (X, d_X) and y_n converges to y in (Y, d_Y) iff and only if (x_n, y_n) converges to (x, y) in $(X \times Y, d_{X \times Y})$

- (a) (5 pts) Show that if F has a compact graph then F is compact-valued.
- (b) (5 pts) Show that the converse may not hold, i.e. F maybe compact-valued but still not have a compact graph.
- (c) (5 pts) Show that if F has a compact graph then F is locally bounded at any $x_0 \in X$.
- (d) (5 pts) If F has a compact graph, is it true that F would be uhc? Explain.